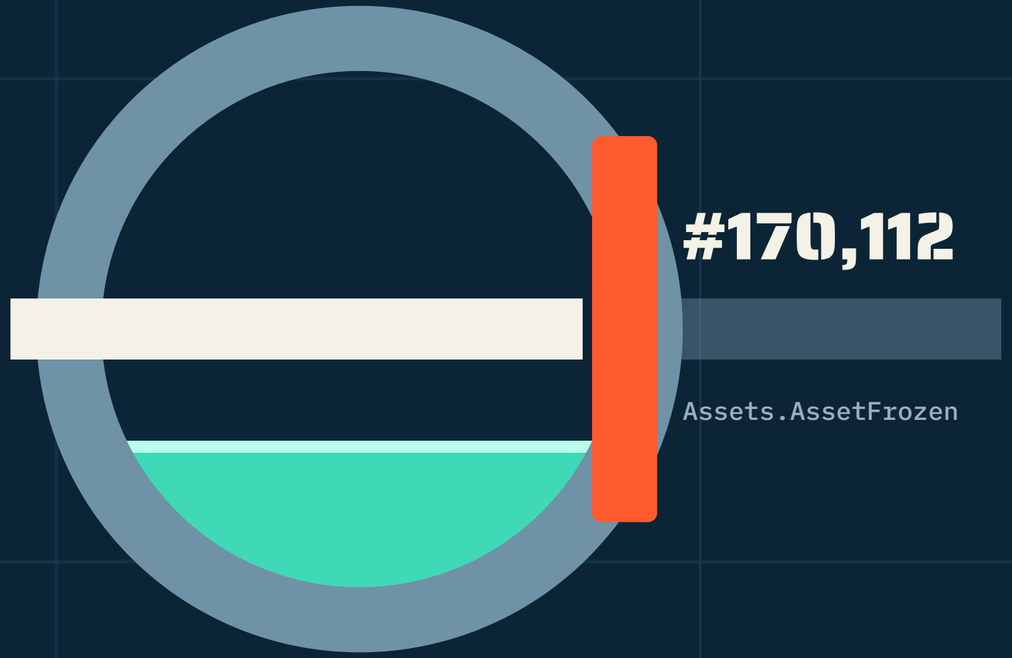
PORTALDOT HACKER HOUSE 2026 · MULTI-CHAIN & CROSS-CHAIN INFRA

Ballast

A reserve-gated circuit breaker for a custodial test dollar on Portaldot.

> The reserve line sinks below the load line — the breaker snaps shut, and pUSDC freezes on Portaldot.



On the Portaldot V3 testnet, 2026-10-24: watcher W2 froze asset 4401 in extrinsic `170112-1`.
Week 2 check-in · `v0.0.1`

RESERVES DRAIN. NOBODY SEES.

As far as we found, Portaldot has no stable unit of account. The realistic first step is a custodial wrapped dollar, and when its reserves drain, every holder keeps trading an IOU that is already worth less than its face value.

01

104.00

pUSDC-test in supply, matched 1:1 by test tokens in the vault.

02

−31.20

The custodian calls `release()`: 30% of the reserve leaves the vault.

03

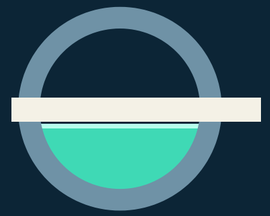
72.80

Reserves left. Supply still says 104.00 unless something on Portaldot acts.

Figures from Ballast's own V3 drill (`evidence/roles-drill-v3-20261024T045513Z.json`; reserve on a local anvil chain, NOT Sepolia). The person it protects: a dApp builder deciding whether checkout may accept pUSDC-test.

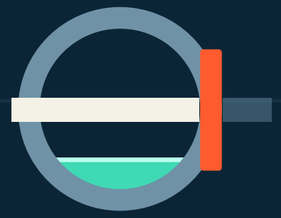
10 09 08 07 06 05 04 03 02 01

FOUR ROLES, THREE MULTISIGS



LEVEL

reserves ≥ supply · asset Live · transfers work



TRIPPED

reserves < supply · asset Frozen · `Assets.transfer` refused

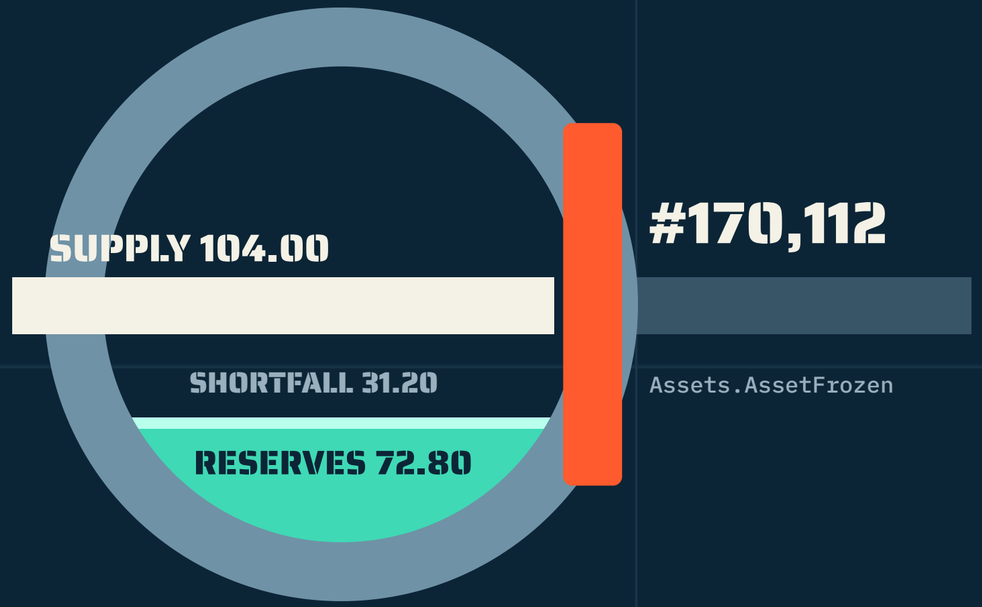
ROLE	HELD BY	WHAT IT MAY DO
FREEZER	`M_watch` 1-of-2 watchers W1, W2	Any one watcher freezes the asset when reserves < supply. Its thaw attempts fail with `Assets.NoPermission`.
ISSUER	`M_mint` 2-of-2 operator A, reserve bot B	The bot co-signs a mint only if *finalized* reserves cover supply plus the amount, and never while frozen.
ADMIN	`M_admin` 2-of-3 A, A2, B	The bot co-signs a thaw only after finality passes the drain block and reserves cover supply again.
RECORD	bot B remarks + EVM mirror	`BALLAST/1` attestations on chain, mirrored into `ReserveOracle`; `ballast verify` re-reads both chains.

Roles are set with `Assets.set_team`; the reserve check itself runs off chain, in the bot and the watchers.

LIVE PROOF · PORTALDOT V3 TESTNET · 2026-10-24

RESERVES SANK. ASSET FROZE.

10 09 08 07 06 05 04 03 02 01



Gauge values = the freeze remark: s=104000000, r=72800000 at 6 decimals. Reserve: local anvil chain, NOT Sepolia. 12.22 s from drain to freeze included (wall clock, one machine).

anvil #1622
RESERVE CHAIN

Custodian drains 30% (31.20) with `release()`.

170112-1
V3 · WATCHER W2

`M_watch` freezes asset 4401 (`Assets.freeze_asset`, Ok). The reason rides in the same extrinsic:

```
BALLAST/1|freeze|a=4401|reason=deficit|s=104000000|r=72800000|p=anvil|eb=1622|eh=0x3987686a…
```

170114-1
V3 · A HOLDER

`Assets.transfer` 1.00 → `Err(Assets.AssetNotLive)`

170115-1
V3 · WATCHER W1

A watcher tries to thaw → `Err(Assets.NoPermission)`. The freezer cannot undo its own freeze.

10 09 08 07 06 05 04 03 02 01

EVERY GATE, SIGNED OR REFUSED

evidence/roles-drill-v3-20261024T045513Z.json · reserve bot B decisions (excerpt; … marks a cut)

```
$ ballast bot cosign-mint 1.00 …  signed
  finalized reserves 104000000 (latest 104000000) >= S1 53000000 + S2 50000000 + amount 1000000 =
104000000
$ ballast bot cosign-mint 0.01 …  refused
  over-mint: finalized reserves 104000000 < S1 54000000 + S2 50000000 + amount 10000 = 104010000
$ ballast bot cosign-mint 0.01 …  refused   (asset frozen)
  I7: the breaker asset is frozen — no mint over a tripped breaker
$ ballast bot cosign-thaw  refused
  finalized block #1559 is older than the deficit block #1622: finality has not reached the drain yet
$ ballast bot cosign-thaw  signed
  finalized reserves 104000000 … >= S1 54000000 + S2 50000000 = 104000000, finalized block #1624 past
the freeze
$ ballast bot cosign-team …  refused   the freezer would change …
$ ballast bot cosign-team …  refused   admin must stay M_admin … another admin could thaw without the
bot …
```

ON CHAIN, V3 TESTNET

170098-1	mint 1.00 executes
170112-1	freeze, asset Frozen
170114-1	transfer refused: `AssetNotLive`
170115-1	watcher thaw refused: `NoPermission`
170133-1	gated thaw, asset Live
170134-1	transfer 1.00 Ok
170156-1	safe `set_team`, co-signed

79/79 V3 extrinsics re-queried, all final ≤ #170291

22/22 role drill steps, 20/20 state checks

15/15 attestation drill steps, 6/6 checks

16/16 role setup steps, 6/6 checks

10 09 08 07 06 05 04 03 02 01

THE RUNTIME ENFORCES IT

solid: Portaldot V3 runtime · dashed: off-chain processes, one key each · bottom: the reserve chain

PORTALDOT V3 RUNTIME · ENFORCEMENT

Assets · asset 4401 pUSDC-test
owner = issuer = `M_mint` · admin = `M_admin` · freezer = `M_watch` · asset 4402: control with keyless roles

System remarks
`BALLAST/1` attest · heartbeat · freeze

Revive · ReserveOracle
`post()` attester only · `status()` · `isBacked(maxAge)` for EVM dApps
`0x4832…155e`

`freeze_asset` | `mint · set_team` | `thaw_asset`

M_watch · 1-of-2
W1, W2 · `as_multi_threshold_1`

M_mint · 2-of-2
A, B · `as_multi`

M_admin · 2-of-3
A, A2, B · `as_multi`

`batch_all[remark, freeze]` | `A: first approval` | `B: final, only if the gate passes` | `attest · heartbeat` | `reads all four`

watchdog
keys W1, W2 · freeze only · reads latest on each provider

operator
keys A, A2 · proposes mint, thaw, team

reserve bot
key B · co-signs mint, thaw, set_team · SQLite decision log

attester
key B · remark first, then `post()` to the oracle after inclusion

ballast verify
re-reads both chains by block hash · rejects wrong signers

`balanceOf @ latest, each provider` | `@ finalized and latest` | `@ finalized` | `by block hash`

Reserve chain · ReserveVault.sol + ERC-20 reserve token
`reserves()` · `release()` owner-only · in these drills: a local anvil chain, NOT Sepolia · target: Sepolia with Circle test USDC

10 09 08 07 06 05 04 03 02 01

THE STOP LIVES IN THE ASSET

QUESTION	A RESERVE NUMBER ON ITS OWN	BALLAST, MEASURED ON V3
WHO STOPS TRANSFERS?	each dApp that chose to read it	the runtime, for every holder: `Assets.transfer` → `AssetNotLive` (`170114-1`)
CAN THE STOPPER RESTART IT?	the number carries no role on the asset	no: a watcher's thaw → `NoPermission` (`170115-1`)
WHEN CAN IT RESTART?	whenever a consumer decides	only after finality passes the drain: refused at #1559 < #1622, thawed in `170133-1`
CAN SOMEONE FAKE THE RECORD?	depends on who signs the feed	a forged remark from another key (`170253-1`) → `ballast verify` REJECTED; 4/4 bot attestations PASS

Built from pallet-native modules (Assets roles, Multisig, System remarks) with no contract code in the enforcement path. The one contract, `ReserveOracle`, is a read-only mirror: it cannot freeze or thaw anything.

ECOSYSTEM FIT

PORTALDOT'S OWN MODULES

10
09
08
07
06
05
04
03
02
01

ASSETS

`create` · `set_metadata` · `set_team` · `mint` · `freeze_asset` · `thaw_asset` · `transfer`

breaker asset 4401 and control asset 4402 on V3

MULTISIG

`as_multi` · `as_multi_threshold_1` · `cancel_as_multi`

three role multisigs: 2-of-2, 2-of-3, 1-of-2

UTILITY

`batch_all`

the freeze and its reason remark land as one extrinsic

SYSTEM

`remark_with_event`

`BALLAST/1` attestations, heartbeats, freeze reasons

REVIVE

`instantiate_with_code` · `call`

`ReserveOracle` on V3 (deployed in 169960-1): any V3 Solidity dApp can ask `isBacked(maxAge)`

PINNED TO THE LIVE CHAIN

GENESIS

`0xd4f9acbc…5283bd`

RUNTIME

`portaldot-v3 / 3002`

BLOCKS

`6000 ms`

OPEN SOURCE

MIT · one CLI, `ballast` · 274 tests (239 pytest, 20 Foundry, 15 Hardhat)

WHO NEEDS A LOAD LINE

WHO IT IS FOR

The first issuer of a dollar on Portaldot, who needs a safety layer it can show, and that the ecosystem can require of any issuer.

The dApp builder deciding whether checkout may accept that dollar, who must tell users why it is safe.

PATH (PLAN, NOT BUILT)

1 **TESTNET, TODAY**
V3 drills pass; one person holds every key; reserve on local anvil.

2 **SEPOLIA RESERVE**
`ReserveVault` with 100.00 test USDC; a watcher on separate infrastructure.

3 **A REAL ISSUER**
watcher and attester keys held by parties other than the issuer; community-watcher mode.

No external users and no revenue yet. The engine is open source; the first goal is one issuer running it.

DONE ON V3. NEXT: THE FINAL

DONE

- Role split + `ReserveOracle` on V3: PASS `16/16 + 6/6`
- Role drill on V3: PASS `22/22 + 20/20`
- Attestation drill on V3: PASS `15/15 + 6/6`
- `79/79` V3 extrinsics final, re-queried by `check_receipts.py --profile v3`
- `274` tests · `130` commits since 2026-10-14

NEXT

- Sepolia `ReserveVault` with 100.00 test USDC (the wallet holds 80.00), then the drill with the real finality wait
- Outage and heartbeat drills on V3 (stand-in only so far)
- A watcher on separate infrastructure
- Watcher-set rotation, `M_admin` rotation, wind-down
- Benchmarks over many drills; a reserves page

Scan for the repository.

REPOSITORY
`github.com/edycutjong/portaldot-hackathon-2026-ballast-edycu`

V3 PROOF TABLES
`docs/PROOF.md`

THE FREEZE, IN THE NODE CONSOLE
`block #170,112`

AI assistance: built with Claude Code (Anthropic). Most of the code, the tests and the prose were drafted with AI assistance and reviewed by the author. The architecture, the role map and the on-chain design decisions are the author's.